The completed Life Table is as follows:

x	$l(x)$	$d(x)$	$p(x)$	$q(x)$	$L(x)$	$T(x)$	$e^0(x)$
72	4412	688	0.8441	0.1559	4,068	34,098	7.73
73	3724	523	0.8596	0.1404	3,463	30,030	8.06
74	3201	642	0.7994	0.2006	2,880	26,567	8.30

Example 8.9

Find the missing values in the following Life Table:

Age (in years)	$l(x)$	$d(x)$	$p(x)$	$q(x)$	$L(x)$	$T(x)$	$e^0(x)$
4	95,000	500	-	-	-	48,50,300	-
5	-	400	-	-	-	-	-

Solution:

Value of the survivorship function $l(x)$ at $x = 5$ years can be estimated as

$$l(5) = l(4) - d(4)$$

$$= 95000 - 500$$

$$= 94500$$

Values of $q(x)$ are estimated as

$$q(4) = \frac{d(4)}{l(4)}$$

$$= \frac{500}{95000}$$

$$= 0.005$$

$$q(5) = \frac{d(5)}{l(5)}$$

$$= \frac{400}{94500}$$

$$= 0.004.$$

Values of $p(x)$ are estimated from the corresponding values of $q(x)$ as

$$p(4) = 1 - q(4)$$

$$= 1 - 0.005 = 0.995$$

$$p(5) = 1 - q(5)$$

$$= 1 - 0.004 = 0.996.$$

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